

Deep Learning and Reinforcement Learning:

Lecture notes

Contents

1	Introduction	1
1.1	A motivating regression example	1
1.2	Population risk, empirical risk, and the empirical measure	2
1.3	Exact ERM and the three-error decomposition	3
1.4	Capacity, underfitting, and overfitting	3
1.5	Double descent	8

Lecture 1

Introduction

1.1 A motivating regression example

Suppose we observe a sample

$$S = \{(X_i, Y_i)\}_{i=1}^n. \quad (1.1.1)$$

A useful motivational model is

$$Y = f^\dagger(X) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid X] = 0. \quad (1.1.2)$$

Here $f^\dagger$ is an unknown signal and ε represents intrinsic randomness or unobserved variation. Gaussian noise is one example, not a universal assumption. Most of the theory below does not require the additive model (1.1.2).

For a predictor $f : \mathcal{X} \rightarrow \mathbb{R}$, the empirical squared loss is

$$\hat{L}_n(f) := \frac{1}{2n} \sum_{i=1}^n (Y_i - f(X_i))^2. \quad (1.1.3)$$

The linear class is

$$\mathcal{F}_{\text{lin}} := \{f_\theta : x \mapsto \theta^\top x, \theta \in \mathbb{R}^d\}. \quad (1.1.4)$$

Choosing this class makes a representation assumption: the signal is being approximated by a linear function. An affine predictor $x \mapsto \theta^\top x + b$ includes an intercept b . This conventional *bias parameter* is unrelated to the statistical bias of an estimator defined later in this chapter.

Generalized linear classes. For a fixed response map $\psi : \mathbb{R} \rightarrow \mathbb{R}$, consider

$$\mathcal{F}_{\text{glm}} := \{f_\theta : x \mapsto \psi(\theta^\top x), \theta \in \mathbb{R}^d\}. \quad (1.1.5)$$

The logistic map $\psi(u) = (1 + e^{-u})^{-1}$ produces probabilities for a binary response, while $\psi(u) = e^u$ produces a positive conditional mean. A full generalized linear model pairs the response map with an observation model and a corresponding loss. The predictor is nonlinear in x after ψ , but it still depends on x only through the one-dimensional score $\theta^\top x$.

Neural-network classes. With the ReLU activation $\rho(u) := \max\{u, 0\}$, a one-hidden-layer class is

$$\mathcal{F}_{\text{NN}}(m) := \left\{ x \mapsto a_0 + \sum_{j=1}^m a_j \rho(w_j^\top x + b_j) : a_0, a_j, b_j \in \mathbb{R}, w_j \in \mathbb{R}^d \right\}. \quad (1.1.6)$$

Unlike the linear and generalized linear classes above, which use the single score $\theta^\top x$ with a fixed response map, a neural network learns the hidden features $x \mapsto \rho(w_j^\top x + b_j)$ together with their output weights. This flexibility usually makes the training objective nonconvex.

These examples separate several kinds of modeling flexibility. A linear predictor uses the identity response map; a generalized linear class adds a fixed nonlinear response map; and a neural network learns multiple hidden features. Moving to a richer class or increasing m can reduce approximation error, but it also changes the statistical and optimization problems. In every case the observations are the labeled sample S , and performance is judged by prediction loss on a fresh draw rather than only by training loss.

1.2 Population risk, empirical risk, and the empirical measure

Let $Z = (X, Y)$ have an unknown distribution P on $\mathcal{X} \times \mathcal{Y}$. A loss $\ell : \mathcal{U} \times \mathcal{Y} \rightarrow [0, \infty)$ evaluates a prediction.

Common choices of loss. The loss must match the prediction space and the response encoding. Common examples include

$$\begin{aligned} \ell_{\text{sq}}(a, y) &:= \frac{1}{2}(a - y)^2, & a, y &\in \mathbb{R}, \\ \ell_{0-1}(a, y) &:= \mathbf{1}\{ya \leq 0\}, & a &\in \mathbb{R}, \quad y \in \{-1, +1\}, \\ \ell_{\text{bce}}(p, y) &:= -y \log p - (1 - y) \log(1 - p), & p &\in (0, 1), \quad y \in \{0, 1\}. \end{aligned}$$

The squared loss is the choice used in (1.1.3); its unrestricted population target is the conditional mean. The 0-1 loss directly counts classification errors but is discontinuous. Binary cross-entropy assesses a predicted probability. For a real-valued score and a signed label, its logistic form is

$$\ell_{\log}(a, y) := \log(1 + e^{-ya}), \quad a \in \mathbb{R}, \quad y \in \{-1, +1\}.$$

For regression with possible outliers, the absolute loss $\ell_{\text{abs}}(a, y) := |a - y|$ is a robust choice whose unrestricted population target is a conditional median. A differentiable compromise is the Huber loss

$$\ell_{\text{H},\delta}(a, y) := \begin{cases} \frac{1}{2}(a - y)^2, & |a - y| \leq \delta, \\ \delta|a - y| - \frac{1}{2}\delta^2, & |a - y| > \delta, \end{cases} \quad \delta > 0.$$

It is quadratic for small residuals and linear in the tails, so a large residual has less influence than under squared loss. Only the 0-1 loss above is automatically bounded by 1; the other losses generally require additional assumptions or different concentration arguments.

Definition 1.1 (Population and empirical risk). The population risk of $f : \mathcal{X} \rightarrow \mathcal{U}$ is

$$L(f) := \mathbb{E}_{Z \sim P}[\ell(f(X), Y)]. \quad (1.2.1)$$

For independent observations $Z_1, \dots, Z_n \sim P$, the empirical risk is

$$\widehat{L}_n(f) := \frac{1}{n} \sum_{i=1}^n \ell(f(X_i), Y_i). \quad (1.2.2)$$

Define the empirical measure

$$P_n := \frac{1}{n} \sum_{i=1}^n \delta_{Z_i}. \quad (1.2.3)$$

Then $L(f) = P\ell_f$ and $\widehat{L}_n(f) = P_n\ell_f$, where $\ell_f(x, y) := \ell(f(x), y)$. Learning replaces an expectation under the unknown P by an expectation under the random P_n .

Quantity	Data role	Interpretation
Training error $\widehat{L}_{\text{tr}}(f)$	Fits ordinary model parameters	An empirical objective, usually optimistically small after fitting
Population or generalization error $L(f)$	Defined by the unknown distribution	Expected loss on a fresh draw; the target quantity
Validation error $\widehat{L}_{\text{val}}(f)$	Selects hyperparameters or stopping time	A selection criterion, not a final unbiased report after repeated use
Test error $\widehat{L}_{\text{test}}(f)$	Used after all choices are frozen	A finite-sample estimate of $L(f)$, with its own uncertainty

1.3 Exact ERM and the three-error decomposition

Given a model class $\mathcal{F}$, empirical risk minimization is the prescription

$$\hat{f}_n \in \arg \min_{f \in \mathcal{F}} \hat{L}_n(f). \quad (1.3.1)$$

The operator $\arg \min$ hides computation. A practical optimizer returns an iterate $\tilde{f}_n$, not an abstract exact minimizer.

Definition 1.2 (Empirical optimization gap). The empirical optimization gap of $\tilde{f}_n \in \mathcal{F}$ is

$$\varepsilon_{\text{opt}} := \hat{L}_n(\tilde{f}_n) - \inf_{f \in \mathcal{F}} \hat{L}_n(f). \quad (1.3.2)$$

Equivalently,

$$\hat{L}_n(\tilde{f}_n) \leq \inf_{f \in \mathcal{F}} \hat{L}_n(f) + \varepsilon_{\text{opt}}. \quad (1.3.3)$$

The gap measures objective value, not distance between parameters. In an overparameterized problem many parameters may have the same empirical loss, so a small gap alone does not identify which solution the algorithm selects.

For the linear regression example, let $\mathbf{X} \in \mathbb{R}^{n \times d}$ have rows $X_i^\top$ and let $\mathbf{y} = (Y_1, \dots, Y_n)^\top$. Exact ERM satisfies the normal equation

$$\mathbf{X}^\top \mathbf{X} \hat{\theta} = \mathbf{X}^\top \mathbf{y}. \quad (1.3.4)$$

If $\mathbf{X}^\top \mathbf{X}$ is invertible, this gives a unique closed form; otherwise minimizers may not be unique. Large neural-network objectives do not offer such a formula, which is why the algorithmic output must be kept separate from the abstract ERM prescription.

Let $\mathcal{F}_{\text{all}}$ be a broad benchmark class and let $\mathcal{F} \subseteq \mathcal{F}_{\text{all}}$ be the working model class. Set

$$L^* := \inf_{f \in \mathcal{F}_{\text{all}}} L(f), \quad L_{\mathcal{F}}^* := \inf_{f \in \mathcal{F}} L(f), \quad \Delta_n(\mathcal{F}) := \sup_{f \in \mathcal{F}} |L(f) - \hat{L}_n(f)|. \quad (1.3.5)$$

The exact split

$$L(\tilde{f}_n) - L^* = \underbrace{L_{\mathcal{F}}^* - L^*}_{\text{approximation}} + \underbrace{L(\tilde{f}_n) - L_{\mathcal{F}}^*}_{\text{within-class excess risk}} \quad (1.3.6)$$

separates the price of restricting the representation from everything that happens within the class. Uniform comparison of L and $\hat{L}_n$ then separates sampling from computation.

Theorem 1.3 (Master excess-risk bound). *If $\tilde{f}_n$ satisfies (1.3.3), then*

$$L(\tilde{f}_n) - L^* \leq \underbrace{L_{\mathcal{F}}^* - L^*}_{\text{approximation error}} + \underbrace{2\Delta_n(\mathcal{F})}_{\text{statistical control}} + \underbrace{\varepsilon_{\text{opt}}}_{\text{optimization error}}. \quad (1.3.7)$$

Proof. For every comparator $f \in \mathcal{F}$,

$$\begin{aligned} L(\tilde{f}_n) &\leq \hat{L}_n(\tilde{f}_n) + \Delta_n(\mathcal{F}) \\ &\leq \inf_{g \in \mathcal{F}} \hat{L}_n(g) + \varepsilon_{\text{opt}} + \Delta_n(\mathcal{F}) \\ &\leq \hat{L}_n(f) + \varepsilon_{\text{opt}} + \Delta_n(\mathcal{F}) \\ &\leq L(f) + \varepsilon_{\text{opt}} + 2\Delta_n(\mathcal{F}). \end{aligned}$$

Subtract L^* and take the infimum over $f \in \mathcal{F}$. □

1.4 Capacity, underfitting, and overfitting

Capacity describes the variety of functions that a learning procedure can effectively choose among [5, Section 5.2]. It is therefore not synonymous with parameter count: two parameterizations may describe the same functions, while a norm constraint or a training rule may make only a small part of a large parameterized family practically accessible.

Representational capacity and three different risks

Suppose that a complexity index q produces nested classes

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \quad (1.4.1)$$

Depending on the application, q might count selected features, polynomial degree, network width or depth, or the radius of a norm constraint.

Concrete example: polynomial degree. For a scalar input, consider

$$\mathcal{F}_q^{\text{poly}} := \left\{ x \mapsto \sum_{j=0}^q \theta_j x^j : \theta \in \mathbb{R}^{q+1} \right\}. \quad (1.4.2)$$

If the signal is approximately quadratic, a line may miss systematic curvature, while a quadratic can capture the main pattern. At the other extreme, with n distinct training inputs, a polynomial of degree at most $n-1$ can interpolate arbitrary observed responses.

The construction in Figure 1.1 follows the book's example: six noiseless observations lie exactly on a quadratic signal. A line cannot express its curvature. The quadratic class contains the signal. A degree-nine class contains the signal too, but its ten coefficients are underdetermined by six interpolation constraints, so many functions achieve zero training loss. A tie-breaking rule such as a Moore–Penrose solution can select an interpolant whose behavior between observations is very different. Thus sparse sampling alone can permit overfitting; label noise would make the problem still more visible.

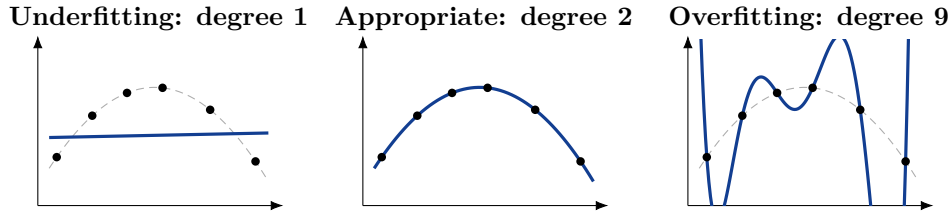


Figure 1.1: Three fits to the same sparse, noiseless quadratic sample. The gray dashed curve is the signal and the blue curve is the fitted predictor. The right panel shows one possible degree-nine interpolant: every observed point is fitted exactly, yet the behavior between points is poorly constrained. Adapted from [5, Figure 5.2].

The example makes capacity visible, but it does not make degree or parameter count a universal measure. Architecture, parameter sharing, norms, and the training algorithm all change which fitted functions are easy to obtain.

Returning to the general nested-class setup, three quantities must be kept separate:

$$\hat{L}_{n,q}^* := \inf_{f \in \mathcal{F}_q} \hat{L}_n(f), \quad L_{\mathcal{F}_q}^* := \inf_{f \in \mathcal{F}_q} L(f), \quad \hat{f}_{q,S} := A_q(S). \quad (1.4.3)$$

The first is the best training risk available in the class, the second is the best population risk available in the class, and the third is what a specified algorithm actually returns from sample S . Nesting gives the deterministic monotonicity relations

$$\hat{L}_{n,q+1}^* \leq \hat{L}_{n,q}^*, \quad L_{\mathcal{F}_{q+1}}^* \leq L_{\mathcal{F}_q}^*. \quad (1.4.4)$$

In contrast, $L(\hat{f}_{q,S})$ need not decrease with q . The fitted predictor depends on a finite sample, initialization, optimization, and any regularization or stopping rule. A larger class creates more opportunities to represent useful structure, but also more opportunities for the procedure to respond to sample-specific variation.

In particular, nesting does not produce a U-shaped population-risk curve: both class-oracle quantities in (1.4.4) decrease.

Bias–variance trade-off under squared loss

Let $S \sim P^n$ be the training sample and let $\hat{f}_{q,S} := A_q(S)$ be the predictor returned at capacity q . We will use the repeated-sample averages

$$T(q) := \mathbb{E}_S [\hat{L}_n(\hat{f}_{q,S})], \quad R(q) := \mathbb{E}_S [L(\hat{f}_{q,S})]. \quad (1.4.5)$$

For the exact additive explanation under squared loss, let $m(x) := \mathbb{E}[Y | X = x]$, let the evaluation pair (X, Y) be independent of S , and set $\bar{f}_q(x) := \mathbb{E}_S[\hat{f}_{q,S}(x)]$. At a fixed input x , the estimator bias and variance are

$$\text{Bias}_S\{\hat{f}_{q,S}(x)\} := \bar{f}_q(x) - m(x), \quad \text{Var}_S\{\hat{f}_{q,S}(x)\} := \mathbb{E}_S[\{\hat{f}_{q,S}(x) - \bar{f}_q(x)\}^2].$$

They are properties of the learning procedure across hypothetical repetitions of the complete training sample, not quantities that can be read from one fitted model. Because this chapter uses $\ell(a, y) = \frac{1}{2}(a - y)^2$,

$$R(q) = \underbrace{\frac{1}{2}\mathbb{E}_X[\text{Var}(Y | X)]}_{\text{irreducible noise}} + \underbrace{\frac{1}{2}\|\bar{f}_q - m\|_{L^2(P_X)}^2}_{\text{squared estimator bias}} + \underbrace{\frac{1}{2}\mathbb{E}_X[\text{Var}_S\{\hat{f}_{q,S}(X)\}]}_{\text{estimator variance}}. \quad (1.4.6)$$

To verify the identity, write

$$Y - \hat{f}_{q,S}(X) = \{Y - m(X)\} + \{m(X) - \bar{f}_q(X)\} + \{\bar{f}_q(X) - \hat{f}_{q,S}(X)\}.$$

The cross terms vanish after conditioning on X and averaging over the independent training sample. In the motivating additive model, $m = f^\dagger$.

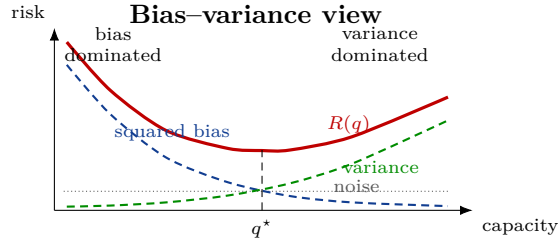


Figure 1.2: Bias–variance view of the finite-sample capacity trade-off under squared loss, adapted from [5, Figure 5.6]. The red curve is the repeated-sample population risk $R(q)$ in (1.4.5), decomposed by (1.4.6). The U shape is a typical schematic, not a universal law.

Diagnosing underfitting and overfitting

Underfitting means that the fitted predictor fails to capture useful, transferable structure. High training and independent-validation losses are a common symptom. The cause may be a restrictive class, incomplete optimization, excessive regularization, inappropriate features, or a poorly matched loss. *Overfitting* means that the fitted predictor uses sample-specific variation in a way that does not transfer. Low training loss together with a substantially larger independent-validation loss is a common symptom. Low training loss by itself is not evidence of overfitting. Likewise, the mnemonics *underfitting means high bias* and *overfitting means high variance* describe common mechanisms, not definitions. Bias and variance average over hypothetical repetitions of the whole training sample; one train–validation split can reveal symptoms but cannot identify either quantity.

Observed pattern	Plausible diagnosis	Useful next comparison
Training and validation losses are both high	The representation or optimization may be too restrictive	Increase class size, train longer, or weaken regularization, changing one factor at a time.
Training loss is low, but validation loss is much higher	The procedure may be sensitive to the sample; leakage or excessive tuning is also possible	Add data or regularization, reduce effective capacity, and audit the split and preprocessing pipeline.
Training and validation losses are both low and close	No obvious underfitting or overfitting signal	Freeze modeling decisions and use the untouched test set once for final estimation.

Learning curves add an important axis to this diagnosis. If the training– validation gap shrinks as the sample size grows, finite-sample variability is a plausible limitation. If both curves plateau at an unsatisfactory level, representation, optimization, regularization, or the chosen loss may be the bottleneck. These patterns are clues, not logical equivalences: distribution shift, label noise, leakage, and a mismatched evaluation metric can imitate the same symptoms.

Hyperparameters, validation, and regularization

Ordinary parameters are fitted inside a chosen training procedure. *Hyperparameters* specify that procedure: examples include class size q , network architecture, a regularization level, learning rate, batch size, and stopping time. Capacity-controlling hyperparameters cannot be chosen by minimizing training error alone, because that criterion systematically favors more fit and less regularization.

A clean protocol is nested:

1. use the training set to produce a candidate $\hat{f}_\lambda$ for each $\lambda \in \Lambda$;
2. use independent validation data to choose $\hat{\lambda}$;
3. freeze every choice and use an untouched test set once to estimate $L(\hat{f}_{\hat{\lambda}})$.

Validation is itself a model-selection problem, so its reported minimum is optimistic after search. If the test set is inspected repeatedly, it has quietly become another validation set. There is no universal 80/20 split: the allocation balances fitting accuracy against selection and evaluation uncertainty. A standard finite-candidate concentration bound can quantify this selection effect.

Regularization expresses a preference among solutions. A constrained version uses

$$\mathcal{F}_R := \{f \in \mathcal{F} : \Omega(f) \leq R\},$$

whereas a penalized procedure minimizes

$$\hat{L}_n(f) + \lambda \Omega(f).$$

The second display defines a different empirical objective for every λ . Its optimization gap must be measured relative to that penalized objective, not silently substituted for ε_{opt} in the unregularized master bound. Regularization can change approximation bias, sampling variability, and optimization geometry at the same time; it is not a fourth independent error term.

Ridge regression as a capacity-control example. For clarity, omit the intercept; in practice it is usually left unpenalized, and all centering constants must be learned from the training set only. For $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$, and $\lambda \geq 0$, ridge regression solves

$$\hat{\theta}_\lambda \in \arg \min_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\theta\|^2 + \frac{\lambda}{2} \|\theta\|^2 \right\}.$$

For $\lambda > 0$, the first-order condition gives

$$\hat{\theta}_\lambda = (\mathbf{X}^\top \mathbf{X} + n\lambda I_d)^{-1} \mathbf{X}^\top \mathbf{y}.$$

The factor $n\lambda$ appears because the empirical squared loss is divided by n . At $\lambda = 0$, take the minimum-norm least-squares solution $\hat{\theta}_0 = \mathbf{X}^\dagger \mathbf{y}$; it is the limit of $\hat{\theta}_\lambda$ as $\lambda \downarrow 0$.

Let $\mathbf{X} = \mathbf{U}_r \text{diag}(s_1, \dots, s_r) \mathbf{V}_r^\top$ be a thin singular-value decomposition and define

$$a_j(\lambda) := \frac{s_j^2}{s_j^2 + n\lambda}.$$

Then

$$\begin{aligned}\hat{\theta}_\lambda &= \sum_{j=1}^r \frac{s_j}{s_j^2 + n\lambda} (\mathbf{u}_j^\top \mathbf{y}) \mathbf{v}_j, \\ \hat{\mathbf{y}}_\lambda &:= \mathbf{X} \hat{\theta}_\lambda = \sum_{j=1}^r a_j(\lambda) (\mathbf{u}_j^\top \mathbf{y}) \mathbf{u}_j.\end{aligned}$$

Thus ridge shrinks the unregularized fit in direction j by $a_j(\lambda)$. Small-singular-value directions are damped most strongly, and the effective degrees of freedom

$$\text{tr}[\mathbf{X}(\mathbf{X}^\top \mathbf{X} + n\lambda \mathbf{I}_d)^{-1} \mathbf{X}^\top] = \sum_{j=1}^r a_j(\lambda)$$

decrease with λ .

The training residual makes clear why training loss cannot choose the amount of regularization:

$$\|\mathbf{y} - \hat{\mathbf{y}}_\lambda\|^2 = \|(I_n - \mathbf{U}_r \mathbf{U}_r^\top) \mathbf{y}\|^2 + \sum_{j=1}^r \left(\frac{n\lambda}{s_j^2 + n\lambda} \right)^2 (\mathbf{u}_j^\top \mathbf{y})^2.$$

Every λ -dependent factor on the right is nondecreasing, so training squared error favors $\lambda = 0$, up to ties. Minimized penalized-objective values should not be used to select λ : changing λ changes the criterion, so their ordering does not estimate a common prediction risk. Instead, fit each candidate on the training set and choose

$$\hat{\lambda} \in \arg \min_{\lambda \in \Lambda} \frac{1}{2n_{\text{val}}} \|\mathbf{y}_{\text{val}} - \mathbf{X}_{\text{val}} \hat{\theta}_\lambda^{\text{tr}}\|^2.$$

Ridge also makes the coefficient bias–variance mechanism explicit. Conditional on the design, suppose $\mathbf{y} = \mathbf{X}\theta^* + \varepsilon$, with $\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0$ and $\text{Cov}(\varepsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$. For $\alpha_j := \mathbf{v}_j^\top \theta^*$,

$$\mathbf{v}_j^\top \hat{\theta}_\lambda = a_j(\lambda) \alpha_j + \frac{s_j}{s_j^2 + n\lambda} \mathbf{u}_j^\top \varepsilon,$$

so the coefficient in direction j has

$$\text{Bias}_j^2(\lambda) = \{1 - a_j(\lambda)\}^2 \alpha_j^2, \quad \text{Var}_j(\lambda) = \frac{\sigma^2 s_j^2}{(s_j^2 + n\lambda)^2}.$$

As λ increases, squared shrinkage bias is nondecreasing and sampling variance is nonincreasing in each observable direction. For isotropic future covariates, summing these terms, together with any unidentifiable null-space bias, gives the prediction bias and variance; a general test covariance weights and couples directions differently. The best trade-off therefore depends on the unknown signal, noise, and future covariate distribution, which is precisely why λ is selected by validation. It also previews why ridge can smooth the interpolation peak discussed next.

Remark 1.4 (k -fold cross-validation). When data are scarce, split the sample into k folds, train k times, and evaluate each observation using a model that did not train on its fold. This uses data efficiently but costs k fits. Because the fitted training sets overlap, fold errors are dependent, so a naive independent-fold standard error is not justified. If cross-validation both selects hyperparameters and reports final performance, an outer evaluation layer or untouched test set is needed.

1.5 Double descent

Figure 1.2 shows the classical picture obtained when capacity is increased through an underfitting regime and into an overfitting regime. If the horizontal axis is continued until the learning procedure first achieves zero training loss, and then continued farther, a second pattern can appear. Write q_{int} for this *interpolation threshold*. The expected population risk may rise sharply near q_{int} and then decrease again in a more strongly overparameterized regime. The resulting curve is called *double descent* [1].

Throughout this subsection, $n := |S|$ is the number of training examples. We hold n fixed while increasing the general capacity index q . In the linear model below we set $q = d$, where d is the number of input coordinates retained by the learner and, because no intercept is fitted, also the number of unknown regression coefficients. The aspect ratio $\gamma := d/n$ distinguishes the underparameterized regime $d < n$, the critical regime $d \approx n$, and the overparameterized regime $d > n$.

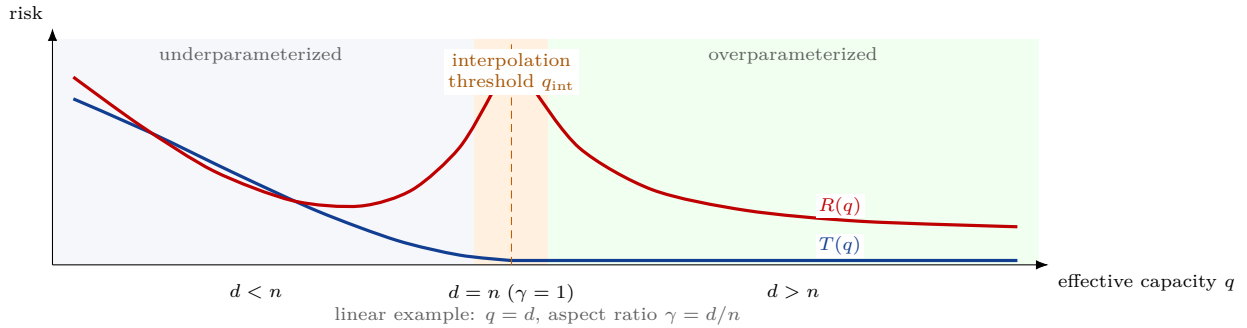


Figure 1.3: Schematic model-wise double descent, adapted from [1]. The quantities $T(q)$ and $R(q)$ are the repeated-sample averages defined in (1.4.5). Training risk first reaches zero at the interpolation threshold. Population risk can peak there and then fall as the procedure moves farther into an overparameterized regime. The shape is qualitative, not a theorem for every nested class; the labels involving d specialize the generic capacity axis to the linear model below.

A Gaussian linear model where the peak is explicit. Let $X = (X_1, X_2, \dots)$ have independent standard Gaussian coordinates and consider the fixed regression function

$$m(X) = \sum_{j \geq 1} \beta_j X_j, \quad Y = m(X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad \varepsilon \perp\!\!\!\perp X,$$

with $\sum_{j \geq 1} \beta_j^2 < \infty$. At capacity d , the learner uses only $X_1, \dots, X_d$ and minimizes squared loss. For a training sample $S = \{(X_i, Y_i)\}_{i=1}^n$, let $\mathbf{X}_d \in \mathbb{R}^{n \times d}$ have row $X_{i,1:d}^\top$, and let $\mathbf{y} = (Y_1, \dots, Y_n)^\top$. Thus $\mathbf{X}_d$ has one row per training example and one column per retained feature.

Solving the three least-squares regimes. For a coefficient vector $b \in \mathbb{R}^d$, the empirical objective, its gradient, and the normal equation are

$$\hat{L}_{n,d}(b) = \frac{1}{2n} \|\mathbf{y} - \mathbf{X}_d b\|_2^2, \quad \nabla \hat{L}_{n,d}(b) = \frac{1}{n} \mathbf{X}_d^\top (\mathbf{X}_d b - \mathbf{y}), \quad \mathbf{X}_d^\top \mathbf{X}_d b = \mathbf{X}_d^\top \mathbf{y}. \quad (1.5.1)$$

A Gaussian n -by- d design has rank $\min\{n, d\}$ almost surely. Consequently, the Moore–Penrose solution has the three explicit forms

$$\hat{\beta}_d := \mathbf{X}_d^\dagger \mathbf{y} = \begin{cases} (\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{X}_d^\top \mathbf{y}, & d < n, \\ \mathbf{X}_d^{-1} \mathbf{y}, & d = n, \\ \mathbf{X}_d^\top (\mathbf{X}_d \mathbf{X}_d^\top)^{-1} \mathbf{y}, & d > n, \end{cases} \quad \hat{f}_d(x) = x_{1:d}^\top \hat{\beta}_d. \quad (1.5.2)$$

For $d < n$, $\mathbf{X}_d^\top \mathbf{X}_d$ is invertible, so the normal equation has the unique ordinary-least-squares solution in the first line. At $d = n$, the square matrix $\mathbf{X}_d$ is invertible and the unique solution fits $\mathbf{X}_d \hat{\beta}_d = \mathbf{y}$. For $d > n$, the matrix has full row rank. Set

$$\beta_0 := \mathbf{X}_d^\top (\mathbf{X}_d \mathbf{X}_d^\top)^{-1} \mathbf{y} = \mathbf{X}_d^\dagger \mathbf{y}.$$

This is one interpolator because

$$\mathbf{X}_d \beta_0 = \mathbf{X}_d \mathbf{X}_d^\top (\mathbf{X}_d \mathbf{X}_d^\top)^{-1} \mathbf{y} = \mathbf{y}.$$

Consequently, if $v \in \ker(\mathbf{X}_d)$, then $\mathbf{X}_d(\beta_0 + v) = \mathbf{y}$, so every $\beta_0 + v$ is an interpolator. Conversely, if b is any interpolator, then $\mathbf{X}_d(b - \beta_0) = \mathbf{y} - \mathbf{y} = 0$, so $b - \beta_0 \in \ker(\mathbf{X}_d)$. The two inclusions prove that the complete solution set is

$$\{b \in \mathbb{R}^d : \mathbf{X}_d b = \mathbf{y}\} = \{\beta_0 + v : v \in \ker(\mathbf{X}_d)\} = \{\mathbf{X}_d^\top (\mathbf{X}_d \mathbf{X}_d^\top)^{-1} \mathbf{y} + v : v \in \ker(\mathbf{X}_d)\}. \quad (1.5.3)$$

Moreover, $\beta_0 \in \text{range}(\mathbf{X}_d^\top)$, while the fundamental theorem of linear algebra gives $\text{range}(\mathbf{X}_d^\top) = \ker(\mathbf{X}_d)^\perp$. Hence $\beta_0 \perp v$ and

$$\|\beta_0 + v\|_2^2 = \|\beta_0\|_2^2 + \|v\|_2^2.$$

Thus the norm is minimized uniquely at $v = 0$, proving that $\mathbf{X}_d^\dagger \mathbf{y}$ is the minimum-norm interpolator. It is also the solution selected by gradient descent from zero for a suitable fixed step size; the zero initialization is essential for this implicit selection.

The distinction is visible directly in the fitted values and training loss. For $d < n$, define the hat matrix $\mathbf{H}_d := \mathbf{X}_d (\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{X}_d^\top$. Then

$$\mathbf{X}_d \hat{\beta}_d = \begin{cases} \mathbf{H}_d \mathbf{y}, & d < n, \\ \mathbf{y}, & d \geq n, \end{cases} \quad \hat{L}_{n,d}(\hat{\beta}_d) = \begin{cases} \frac{1}{2n} \|(\mathbf{I}_n - \mathbf{H}_d) \mathbf{y}\|_2^2, & d < n, \\ 0, & d \geq n. \end{cases} \quad (1.5.4)$$

With nonzero continuous observation or omitted-coordinate noise, the residual in the first line is nonzero almost surely. Thus $d = n$ is the first dimension at which arbitrary training responses can be interpolated. For this specialization $q = d$, the interpolation threshold is $q_{\text{int}} = n$, equivalently $\gamma = 1$.

Define the omitted-signal energy, included-signal energy, and effective noise by

$$a_d := \sum_{j>d} \beta_j^2, \quad B_d^2 := \sum_{j \leq d} \beta_j^2, \quad \tau_d^2 := \sigma^2 + a_d.$$

The omitted coordinates are independent of the fitted coordinates and act like additional Gaussian training noise. More precisely,

$$\mathbf{y} = \mathbf{X}_d \beta_{1:d} + \boldsymbol{\eta}_d, \quad \boldsymbol{\eta}_d \sim \mathcal{N}(0, \tau_d^2 \mathbf{I}_n), \quad \boldsymbol{\eta}_d \perp \mathbf{X}_d.$$

For $d < n$, equation (1.5.4) and $(\mathbf{I}_n - \mathbf{H}_d) \mathbf{X}_d = 0$ give the following calculation:

$$\mathbb{E}[\|(\mathbf{I}_n - \mathbf{H}_d) \mathbf{y}\|_2^2 \mid \mathbf{X}_d] = \tau_d^2 (n - d).$$

This follows because $\mathbf{H}_d$ is a rank- d orthogonal projector; the residual projector has trace $n - d$. Therefore the repeated-sample training curve is exactly

$$T(d) = \mathbb{E}_S [\hat{L}_{n,d}(\hat{\beta}_d)] = \begin{cases} \frac{\tau_d^2}{2} \left(1 - \frac{d}{n}\right), & d < n, \\ 0, & d \geq n. \end{cases} \quad (1.5.5)$$

Deriving the test curve. Set $\theta_d := \beta_{1:d} \in \mathbb{R}^d$, so $\|\theta_d\|_2^2 = B_d^2$. For an independent test observation, write

$$Y' = (X'_{1:d})^\top \theta_d + \eta'_d, \quad \eta'_d := \sum_{j>d} \beta_j X'_j + \varepsilon'.$$

Then $X'_{1:d} \sim \mathcal{N}(0, \mathbf{I}_d)$, $\eta'_d \sim \mathcal{N}(0, \tau_d^2)$, and these variables are independent of each other and of the training sample. With $\Delta_d := \hat{\beta}_d - \theta_d$, conditioning on the realized training sample gives

$$\mathbb{E}[(Y' - \hat{f}_d(X'))^2 \mid S] = \mathbb{E}[(\eta'_d - (X'_{1:d})^\top \Delta_d)^2 \mid S]$$

$$\begin{aligned}
 &= \tau_d^2 + \Delta_d^\top \mathbb{E}[X'_{1:d}(X'_{1:d})^\top] \Delta_d \\
 &= \tau_d^2 + \|\Delta_d\|_2^2.
 \end{aligned}$$

The cross term in the second line is zero because η'_d and $X'_{1:d}$ are independent and centered. Thus it remains to calculate the mean squared coefficient error.

For $d < n$, ordinary least squares gives

$$\Delta_d = (\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{X}_d^\top \boldsymbol{\eta}_d.$$

Consequently,

$$\begin{aligned}
 \mathbb{E}[\|\Delta_d\|_2^2 \mid \mathbf{X}_d] &= \text{tr}(\mathbb{E}[\Delta_d \Delta_d^\top \mid \mathbf{X}_d]) \\
 &= \tau_d^2 \text{tr}((\mathbf{X}_d^\top \mathbf{X}_d)^{-1} \mathbf{X}_d^\top \mathbf{X}_d (\mathbf{X}_d^\top \mathbf{X}_d)^{-1}) \\
 &= \tau_d^2 \text{tr}((\mathbf{X}_d^\top \mathbf{X}_d)^{-1}).
 \end{aligned}$$

Here $\mathbf{X}_d^\top \mathbf{X}_d$ is a d -dimensional Wishart matrix with n degrees of freedom. Its inverse first moment is

$$\mathbb{E}[(\mathbf{X}_d^\top \mathbf{X}_d)^{-1}] = \frac{\mathbf{I}_d}{n - d - 1}, \quad n > d + 1.$$

It follows that

$$\mathbb{E}[\|\Delta_d\|_2^2] = \tau_d^2 \frac{d}{n - d - 1}, \quad d < n - 1.$$

For $d > n$, introduce the d -by- d row-space projector

$$\boldsymbol{\Pi}_d := \mathbf{X}_d^\top (\mathbf{X}_d \mathbf{X}_d^\top)^{-1} \mathbf{X}_d$$

and $\mathbf{A}_d := \mathbf{X}_d^\top (\mathbf{X}_d \mathbf{X}_d^\top)^{-1}$. The minimum-norm solution satisfies

$$\hat{\beta}_d = \boldsymbol{\Pi}_d \theta_d + \mathbf{A}_d \boldsymbol{\eta}_d, \quad \Delta_d = -(\mathbf{I}_d - \boldsymbol{\Pi}_d) \theta_d + \mathbf{A}_d \boldsymbol{\eta}_d.$$

The first term lies in $\ker(\mathbf{X}_d)$, whereas the second lies in $\text{range}(\mathbf{X}_d^\top) = \ker(\mathbf{X}_d)^\perp$. They are therefore orthogonal for every realization. Moreover, $\mathbf{A}_d^\top \mathbf{A}_d = (\mathbf{X}_d \mathbf{X}_d^\top)^{-1}$, so

$$\mathbb{E}[\|\Delta_d\|_2^2 \mid \mathbf{X}_d] = \|(\mathbf{I}_d - \boldsymbol{\Pi}_d) \theta_d\|_2^2 + \tau_d^2 \text{tr}((\mathbf{X}_d \mathbf{X}_d^\top)^{-1}).$$

The Gaussian row space is a uniformly random n -dimensional subspace of $\mathbb{R}^d$. Rotational invariance therefore gives $\mathbb{E}[\boldsymbol{\Pi}_d] = c \mathbf{I}_d$; taking traces and using $\text{tr}(\boldsymbol{\Pi}_d) = n$ gives $c = n/d$. Since $\mathbf{I}_d - \boldsymbol{\Pi}_d$ is itself an orthogonal projector,

$$\mathbb{E}[\|(\mathbf{I}_d - \boldsymbol{\Pi}_d) \theta_d\|_2^2] = \theta_d^\top (\mathbf{I}_d - \mathbb{E}[\boldsymbol{\Pi}_d]) \theta_d = \left(1 - \frac{n}{d}\right) B_d^2.$$

Also $\mathbf{X}_d \mathbf{X}_d^\top$ is an n -dimensional Wishart matrix with d degrees of freedom, and hence

$$\mathbb{E}[(\mathbf{X}_d \mathbf{X}_d^\top)^{-1}] = \frac{\mathbf{I}_n}{d - n - 1}, \quad d > n + 1.$$

Therefore

$$\mathbb{E}[\|\Delta_d\|_2^2] = \left(1 - \frac{n}{d}\right) B_d^2 + \tau_d^2 \frac{n}{d - n - 1}, \quad d > n + 1.$$

Both Wishart identities follow from one inverse-Gram calculation. If $\mathbf{Z} \in \mathbb{R}^{m \times p}$ has independent standard Gaussian entries and $m > p + 1$, then

$$\mathbb{E}[(\mathbf{Z}^\top \mathbf{Z})^{-1}] = \frac{\mathbf{I}_p}{m - p - 1}.$$

Applying it to $\mathbf{Z} = \mathbf{X}_d$ and $\mathbf{Z} = \mathbf{X}_d^\top$ yields the two moments used above and explains the strict dimension conditions.

Combining the coefficient errors with the independent test-noise term τ_d^2 , the expected test mean-squared error is, away from the critical dimensions,

$$\mathcal{E}_d := \mathbb{E}[(Y' - \hat{f}_d(X'))^2] = \begin{cases} \tau_d^2 \left(1 + \frac{d}{n-d-1}\right), & d < n-1, \\ B_d^2 \left(1 - \frac{n}{d}\right) + \tau_d^2 \left(1 + \frac{n}{d-n-1}\right), & d > n+1. \end{cases} \quad (1.5.6)$$

Under the loss convention of this chapter, $R(d) = \frac{1}{2}\mathcal{E}_d$. The expectation in (1.5.6) is not finite at $d \in \{n-1, n, n+1\}$ when $\tau_d^2 > 0$: individual fitted risks are finite almost surely, but nearly singular designs give the average a heavy tail [6].

When does this formula actually give double descent? Equation (1.5.6) permits double descent, but does not force it. On the underparameterized branch,

$$\mathcal{E}_d = \tau_d^2 \frac{n-1}{n-d-1}, \quad \tau_{d+1}^2 = \tau_d^2 - \beta_{d+1}^2.$$

For $d < n-2$, so that both adjacent expectations are finite, direct subtraction gives

$$\begin{aligned} \mathcal{E}_{d+1} - \mathcal{E}_d &= \frac{n-1}{n-d-2}(\tau_d^2 - \beta_{d+1}^2) - \frac{n-1}{n-d-1}\tau_d^2 \\ &= \frac{n-1}{(n-d-2)(n-d-1)} [\tau_d^2 - (n-d-1)\beta_{d+1}^2]. \end{aligned}$$

Consequently,

$$\mathcal{E}_{d+1} < \mathcal{E}_d \iff \beta_{d+1}^2 > \frac{\tau_d^2}{n-d-1}.$$

The reverse inequality gives an ascent, and equality gives a flat step. Thus the decrease in τ_d^2 and the increase in $(n-1)/(n-d-1)$ compete; monotonicity of the two factors does not determine their product. For example, if $\beta_{d+1} = 0$ and $\tau_d^2 > 0$, that step strictly increases risk. If all added coordinates carry no signal, the entire finite underparameterized branch increases, so there is no first descent. Irregularly ordered coefficient magnitudes can even create several local turns.

The comparison from $d = n-2$ to $d = n-1$ is excluded from the finite difference above. When $\tau_{n-1}^2 > 0$, the expected risk at $d = n-1$ is infinite. This produces the singular peak around interpolation, but does not imply that the finite branch $d < n-1$ contains both a descent and an ascent.

On the overparameterized branch, write

$$\mathcal{E}_d = B_d^2 \left(1 - \frac{n}{d}\right) + \tau_d^2 v_d, \quad v_d := \frac{d-1}{d-n-1}, \quad d > n+1.$$

The noise multiplier decreases because

$$v_{d+1} - v_d = \frac{d}{d-n} - \frac{d-1}{d-n-1} = -\frac{n}{(d-n)(d-n-1)} < 0.$$

Since τ_d^2 is also nonincreasing, the effective-noise contribution $\tau_d^2 v_d$ decreases beyond the threshold. In contrast, $B_d^2(1 - n/d)$ is nondecreasing and can offset that reduction. The familiar double-descent shape therefore requires three comparisons to go the right way: informative early coordinates create a first descent, positive effective noise creates the interpolation peak, and post-threshold noise reduction dominates the growing null-space signal term. The formula displays this mechanism; it is not a universal shape theorem.

The three parts of the curve now have distinct causes.

1. As informative coordinates are added, a_d decreases. If that approximation gain initially dominates estimation variability, the first descent appears.
2. Near $d = n$, the smallest singular values of $\mathbf{X}_d$ can be tiny. Least squares divides by those singular values, strongly amplifying label noise and omitted-feature noise. This produces the interpolation peak.
3. For $d > n$, training loss is zero and there are infinitely many interpolators. The minimum-norm selector has noise-amplification term $\tau_d^2 n/(d-n-1)$, which decreases beyond the threshold and can produce the second descent.

Connection to bias and variance. Equation (1.4.6) remains exact. For example, in the overparameterized Gaussian model above, the squared estimator bias and variance contributions to excess prediction MSE are

$$\begin{aligned}\text{Bias}_d^2 &= a_d + \left(1 - \frac{n}{d}\right)^2 B_d^2, \\ \text{Var}_d &= \frac{n}{d} \left(1 - \frac{n}{d}\right) B_d^2 + \tau_d^2 \frac{n}{d - n - 1}.\end{aligned}$$

Their sum is $\mathcal{E}_d - \sigma^2$, up to the factor 1/2 in our loss. Thus double descent is not a new term outside bias–variance; it is a possible nonmonotone shape of their sum. In particular, variance can peak near interpolation and then decrease.

Related literatures. The mechanism is analytically visible in ridgeless or minimum-norm linear regression with random design [6]. Random-feature regression, equivalently a two-layer network whose random first-layer weights are frozen and whose output layer is fit by least squares, provides a nonlinear model with a precise double-descent analysis [10]. Model-wise and epoch-wise double descent have also been observed empirically for CNNs, ResNets, and Transformers trained close to interpolation [12]. These examples are not a sufficient condition: the peak depends on the capacity path, noise, regularization, early stopping, and the optimizer’s implicit choice among interpolators. Ridge regularization or early stopping can substantially smooth or remove it.